

CHAINATHAN SANTHANAM SUDHAKAR

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Education

Masters in Computer Science, Khoury College, Northeastern University, MA, **GPA: 4.0/4.0** Exp Grad: Dec 2025

Bachelors in Computer Science, Vellore Institute of Technology, India, **GPA: 3.64/4.0** Aug 2018 – Jun 2022

Relevant Courses: Foundations of AI, PRCV, Robotic Sensing & Navigation, Robotic Systems, NLP, Applied Statistics

Experience

Quantitative Neuroscience Lab

Boston, MA

Computer Vision Engineer, Research Assistant

Jan 2025 – Mar 2025

- Trained SAM2 and UNet++ models to track protein growth dynamics from large scale fluorescent images datasets.
- Built postprocessing pipeline using KymoButler (pretrained UNet) with artifact removal, and piecewise linear fitting.
- Developed a Python + MATLAB desktop app for interactive analysis and editing of Neurons (high-density graphs).

Computer Vision Engineer, Intern

Jun 2024 – Dec 2024

- Trained Gabor-UNet, Retina-UNet, and ISNet models for centerline segmentation (78% Dice, 91% IoU).
- Created custom data augmentations (patch sampling, noise injection) for low-SNR 16-bit microscopy datasets.
- Developed a video stabilization algorithm for fluorescent neuron dendrite videos.
- Deployed models using AWS SageMaker, tracked jobs with MLflow, and automated evaluations via GitHub Actions.
- Reduced dendrite analysis time from 3 days to 20 minutes through end-to-end pipeline automation.

Societe Generale

Bangalore, India

Data Engineer

Jul 2022 – Aug 2023

- Designed and deployed over 3 enterprise-grade applications on AWS with Docker and Kubernetes.
- Built ETL workflows using SQL to automate ingestion of SaaS data into internal platforms.
- Delivered a secure transaction system handling 1M+ daily records, with AWS S3 bookkeeping.
- Set up real-time monitoring dashboards with Elasticsearch and Kibana, reducing incident response time by 30%.
- Migrated over 50 legacy Bash scripts to modular Airflow DAGs.

Software Engineer Intern

Jan 2022 – Jun 2022

- Led a team to build a Dashboard for KPI analysis using React, doubling middleware productivity.
- Built Python and Spring Boot apps for workflow automation using Datalake and SCD.
- Created an Automated Mailbox System using Azure GraphAPI for time-sensitive email processing.

Vellore Institute of Technology

Amaravati, India

HPC Research Assistant

Aug 2021 – Dec 2021

- Developed fault-tolerant consensus in MPI-based HPC clusters with UNIX failure injection.
- Integrated gossip-based failure tracking with dimensionality reduction for low-latency updates.
- Improved synchronization by 65% and reduced system overhead by 40% across active nodes.

Projects

View Synthesis - Video to 3D Face Model Reconstruction | [Instant-NGP](#), [NeRF](#), [COLMAP](#), [3D Reconstruction](#), [SfM](#)

- Built a 3D face reconstruction pipeline using COLMAP for pose estimation and Instant-NGP for NeRF-based rendering.
- Trained NeRF models on monocular face videos and extracted high-fidelity 3D meshes using marching cubes.
- Applied mesh denoising and format conversion with Open3D to prepare outputs for downstream 3D applications.

Generative Design Module using Diffusion (3D CAD) | [Diffusion](#), [Transformers](#), [Structured Latent Geometry](#), [B-Rep](#) [Demo]

- Built a generative model to create CAD-compatible part designs satisfying geometric and physical constraints.
- Fine-tuned BrepGen, a diffusion transformer with multi-stage diffusion processes for surfaces, edges, and vertices.
- Achieved 84.6% COV, 0.89 MMD, and 0.82 JSD on unconditional generation using the ABC dataset.

Text to Music Generation using Image Diffusion | [Stable Diffusion](#), [PEFT](#), [AWS](#), [Streamlit](#), [LORA](#) [Code]

- Built real-time music generation by converting text to spectrograms images and reconstructing audio from images.
- Achieved 3× speedup via LoRA PEFT on parallel GPUs for Stable Diffusion v1-5.
- Reduced checkpoint size from 7GB to 550MB, saving 14× storage per run in AWS S3.

MRI Synthesis & Segmentation | [Diffusion, UNet, DDPM](#) [\[Code\]](#)

- Built a diffusion-based image translation model (T1→T2) using cross-attention, achieving 23.8 PSNR and 0.81 SSIM.
- Added timestep-aware conditioning to improve convergence speed by 25% over baseline U-Net.
- Boosted reconstruction fidelity by 15% over vanilla DDPM using cross-modality attention and bidirectional training.

Unguided Image Generation | [Diffusion Models, Custom PyTorch Framework](#) [\[Code\]](#)

- Implemented VAE, U-Net and noise samplers like DDPM, DDIM, PGDM from scratch.
- Integrated Karras pre-conditioning for faster convergence, achieving a 4.96 FID score on LSUN bedrooms dataset.

AI-Powered Sketch Drawing Robot | [Robotics, Diffusion Models, ROS](#) [\[Demo\]](#) [\[Code\]](#)

- Generated sketches using Stable Diffusion and contour-based stroke extraction.
- Reduced stroke trajectory length by 40% for faster execution on a 6-DOF robotic arm.
- Achieved $\pm 1\text{mm}$ accuracy via AprilTag-based real-time calibration.

Privacy-Preserving Mobile App (PII in Images & Text) | [YOLO, SAM2, LLM, OCR, Flutter](#) [\[Demo\]](#) [\[Code\]](#)

- Built an on-device app to detect and redact PII like screens, license plates, ID cards, and sensitive text.
- Achieved 92% segmentation accuracy with <550ms latency using finetuned YOLOv11 on edge devices.
- Integrated MobileBERT + Google ML Kit for text redaction on sensitive text in documents.

Annotation of Object Positions (Tracking) in Rapid Motion Videos | [Vision Transformers, MediaPipe](#) [\[Code\]](#)

- Automated object detection, hand tracking and annotation in rapid motion videos.
- Improved accuracy by 35% using ViCLIP + DETR for object and hand detection.

Research Paper Summarization | [Gemma, LLaMA, Mistral, QLoRA](#) [\[Code\]](#)

- Fine-tuned Gemma-7B, LLaMA-7B, and Mistral-7B on ArXiv ML papers, focusing on advanced data processing.
- Leveraged RoPE scaling and QLoRA with half-precision enabling 2.4x batch size while saving 50% GPU memory.
- Deployed LLM with Continuous Batching, and Token Streaming by using Hugging Face's TGI.

GPT for Custom Style Text Generation | [Transformers, GPT, PyTorch](#) [\[Code\]](#)

- Built a GPT model to generate text in custom styles with support for BPE, word-level, and char-level tokenization.
- Implemented the full Transformer architecture from scratch using PyTorch.
- Integrated flexible callback functions for gradient statistics, metric tracking, and debugging ease.

Technical Skills

Languages: Python, Java, JavaScript, MATLAB, MPI

Frameworks: PyTorch, OpenCV, HuggingFace Transformers, Streamlit, Mathematica, ROS, Flutter

MLOps & Tools: MLFlow, AWS, DagsHub, GitHub Actions, Docker, Kubernetes

Specializations: Computer Vision, Diffusion, 3D Reconstruction, View Synthesis, Robotics, SLAM, SfM, NeRFs

Generative AI: Image Inpainting, Image Generation, Audio Generation, Style Transfer, Text Generation, Summarization

Accomplishments

- HPC and AI Technical Workshop Certification, 2020.
- 3rd place in TechEden Hackathon by Google Developers club for building Secure File Transfer application.
- 2nd place in a Microsoft-sponsored Hackathon for developing an Infotainment Mobile Application.
- 2nd place in Engineering Hackathon conducted by VIT for building a Gesture to Speech Smart Gloves for Mute People.